

Math Primer

(11. Optimization I)

Contents



- ▶ Optimization in Finance
- ▶ Types of Optimization
- ▶ Local vs Global extremum
- ▶ Linear Programming
- ▶ Sensitivity Analysis
- ▶ Primal vs Dual
- ▶ Binary Search
- ▶ Newton Raphson



JL Lagrange
(1736-1813)

Optimization for Finance



- ▶ We run into the need for optimization in all our decision-making process which involves maximizing or minimizing entity under resource constraints (budget). All scientific disciplines use optimization for something or the other. Finance is no different !
- ▶ To highlight a few important use cases in Finance that require optimization
 - Utility Maximization (Economics)
 - Portfolio Selection & Asset Allocation (Quantitative Portfolio Management)
 - Fitting Time Series Models on Financial Data (Econometrics)
 - No Arbitrage Model Calibration to fit market prices (Calibration)
 - Balance Sheet Optimization (ALM)
 - And many more ...

Types of Optimization Problems



Optimization Problem

- ▶ A generic definition of optimization problem would be

Given $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ and a set $S \subset \mathbb{R}^n$, find x^* that solves

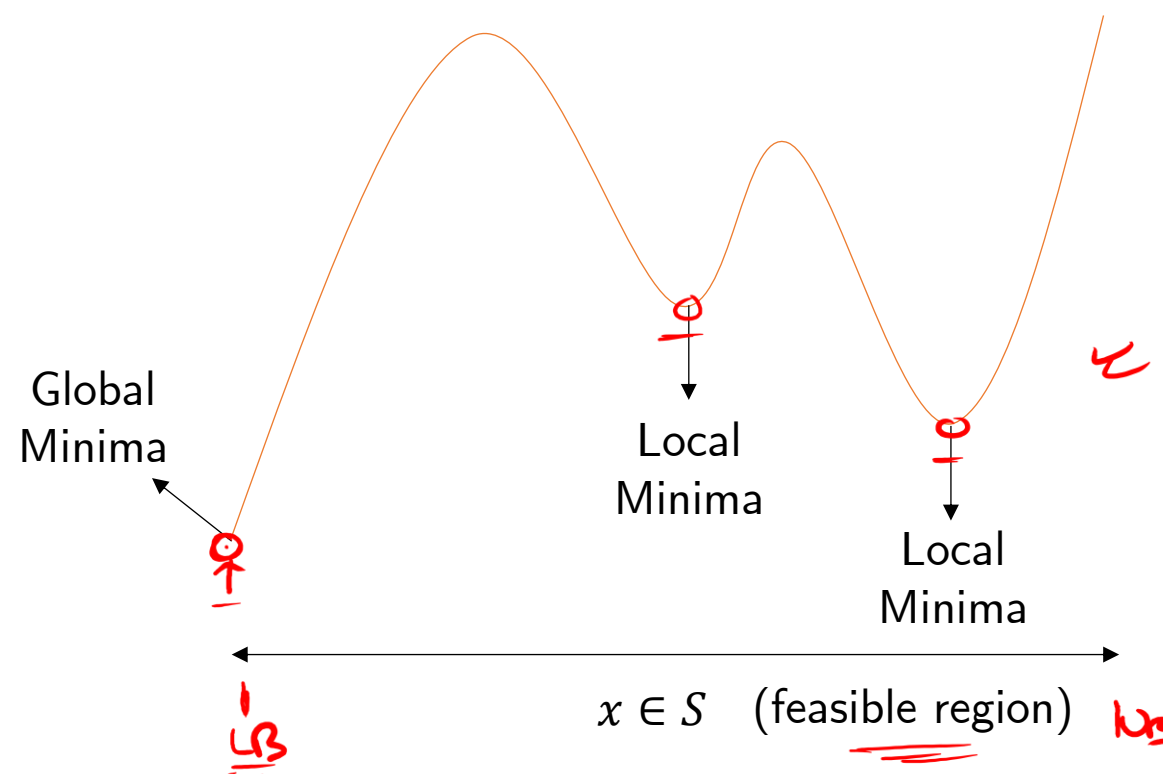
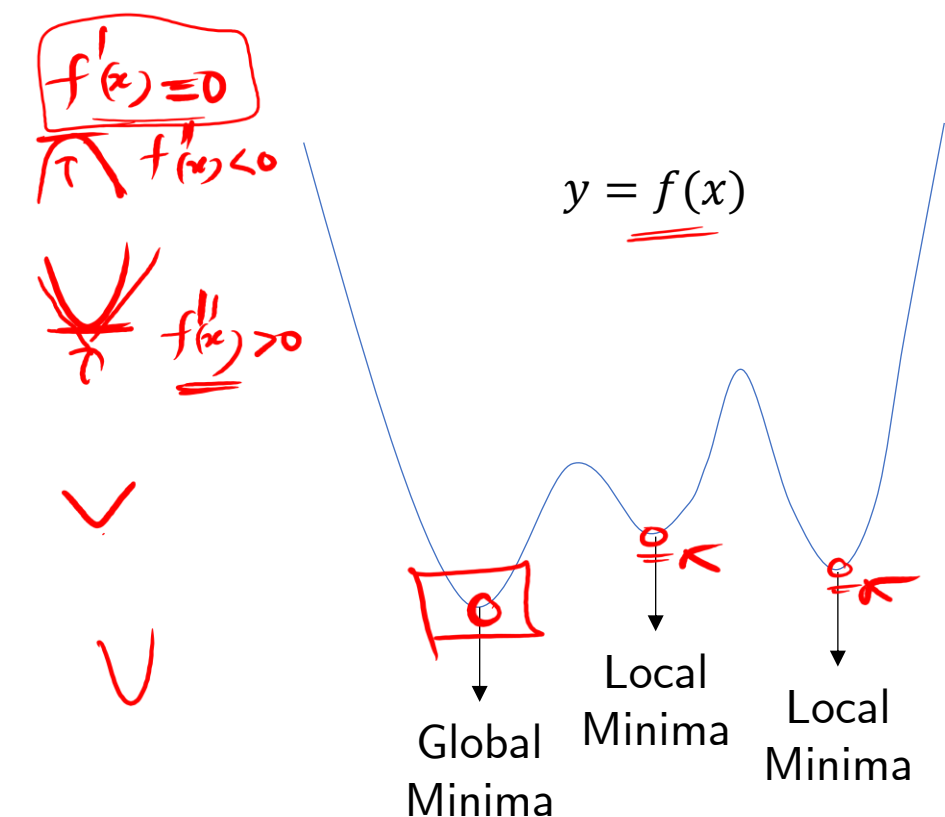
$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & x \in S \end{array}$$

$f =$ Objective function
 $x =$ decision variables
 $S =$ feasible region

- ▶ If we find a solution x^* such that $f(x^*) < f(x)$, $\forall x \in S$, then x^* is the solution to the OP.
- ▶ $f(x^*)$ is called the 'Global Minimum'

Local Minima vs Global Minima

- ▶ A function can have only one global minima but multiple local minima
- ▶ The function must flatten at all local minima, but it may not be flat at global minima within the feasible region

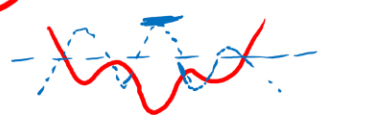


Linear Programming

- An Optimization problem with linear Objective and linear constraints is called a Linear Programming (LP) problem

$$\begin{aligned} \min f(x) \\ = \max -f(x) \end{aligned}$$

x^*



$$\min_x k^T x$$

$$\rightarrow Ax \geq b$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \geq \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad - \quad \begin{cases} 2x_1 + x_2 \geq 2 \\ 3x_1 + 5x_2 \geq 3 \end{cases}$$

$$x_1, x_2$$

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix}^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= (k_1, k_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= k_1 x_1 + k_2 x_2$$

$$\begin{pmatrix} 1 \\ 5 \end{pmatrix}^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 4x_1 + 5x_2$$

- Company XYZ manufactures 2 types of products, A and B. Profit per unit of A is \$3 and per unit of B is \$5. One unit of A takes 1 hour, and one unit of B takes 3 hours to process. Total processing time available is 60 hours. One unit of A needs 3 workers, and one unit of B needs 4 workers. Total workers available are 120. Company has a mandate to manufacture at least 10 units of A. How many units of each the company should make to maximize the profit?

LP Problem (Primal)

Objective Function :

$$\text{Max } z = 3x + 5y$$

Constraints

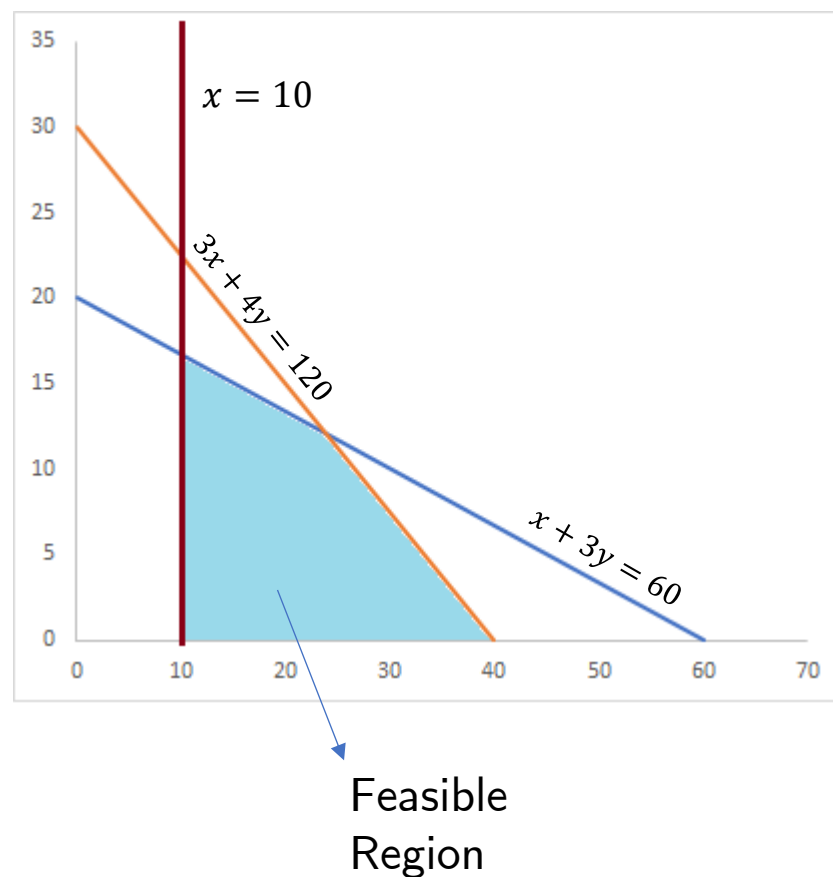
$$x + 3y \leq 60$$

$$3x + 4y \leq 120$$

$$x \geq 10$$

$$x \geq 0$$

$$y \geq 0$$



Optimal Solution : (24,12)

Binding Constraints

$$x + 3y \leq 60$$

$$3x + 4y \leq 120$$

No
slack

Non-binding Constraints

$$x \geq 10$$

surplus = 14

Try objective function:

$$x+5y$$

$$5x+y$$

Scan the corner points of the feasible regions for optimal solution.

Sensitivity Report



- ▶ **Reduced Cost** – How much the objective coefficient of a DV to be reduced for the variable to obtain a positive value in the optimal solution. If final value is already positive, reduced cost = 0
- ▶ **Shadow Price** – Change in Objective score when the constraint boundary is relaxed by 1 unit
- ▶ **Allowable Increase/Decrease** – Maximum allowed change in DV or Constraint boundary that does not alter the solution

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$K\$3 x		24	0	3	0.75	1.333333333
\$K\$4 y		12	0	5	4	1

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$L\$12	$x + 3y \leq 60$ LHS	60	0.6	60	17.5	20
\$L\$13	$3x + 4y \leq 120$ LHS	120	0.8	120	60	23.33333333
\$L\$14	$-x \leq -10$ LHS	-24	0	-10	1E+30	14

LP Problem (Dual)

Primal

$$\boxed{\max} : k^T x$$

$$\underline{Ax \leq b}$$

Max $3x + 5y$

Constraints

$$\begin{aligned} \rightarrow & \quad x + 3y \leq 60 \\ \rightarrow & \quad 3x + 4y \leq 120 \\ & \quad -x \leq -10 \end{aligned}$$

$$\underline{\underline{\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 24 \\ 12 \end{pmatrix}}}$$

Decision variables $\rightarrow$ Constraints
Constraints $\rightarrow$ Decision Variables
 Minima $\rightarrow$ Maxima

Dual

$$\underline{\min} : y^T b$$

$$A^T y \geq k$$

$\boxed{\text{Min}} \quad 60c_1 + 120c_2 - 10c_3$

Constraints

$$\begin{aligned} \rightarrow & \quad c_1 + 3c_2 - c_3 \geq 3 \\ \rightarrow & \quad \underline{3c_1 + 4c_2} \geq \underline{5} \end{aligned}$$

$$\begin{aligned} \rightarrow & \quad \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.8 \\ 0 \end{pmatrix} \end{aligned}$$

Shadow prices

Non-binding constraint

A Short-term Financing Problem



A company has the following cashflows

Month	Jan	Feb	Mar	Apr	May	June
Net Cashflow	-150	-100	200	-200	50	300

The company has access to the following source of funds

- A line of credit of up to \$100K at an interest rate of 1% per month,
- In any one of the first three months, it can issue 90-day commercial paper bearing a total interest of 2% for the 3-month period,
- Excess funds can be invested at an interest rate of 0.3% per month.

The goal is to maximize the value.

A Short-term Financing Problem



Month	Jan	Feb	Mar	Apr	May	June
Net Cashflow	-150	-100	200	-200	50	300
Line of Credit Inflow ✓	$+x_1$	$+x_2$	$+x_3$	$+x_4$	$+x_5$	
Line of Credit Outflow ✓		$-1.01x_1$	$-1.01x_2$	$-1.01x_3$	$-1.01x_4$	$-1.01x_5$
Commercial Paper Inflow	$+y_1$	$+y_2$	$+y_3$	.	.	.
Commercial Paper Outflow				$-1.02y_1$	$-1.02y_2$	$-1.02y_3$
Excess Funds	z_1	z_2	z_3	z_4	z_5	v
Interest on Excess Funds		$+1.003z_1$	$+1.003z_2$	$+1.003z_3$	$+1.003z_4$	$+1.003z_5$

LPP (Short-term Financing)

$$\boxed{\max v}$$

$$x_1 + y_1 - z_1 = 150$$

$$x_2 + y_2 - 1.01x_1 + 1.003z_1 - z_2 = 100$$

$$x_3 + y_3 - 1.01x_2 + 1.003z_2 - z_3 = -200$$

$$x_4 - 1.01x_3 - 1.02y_1 + 1.003z_3 - z_4 = 200$$

$$x_5 - 1.01x_4 - 1.02y_2 + 1.003z_4 - z_5 = -50$$

$$-1.01x_5 - 1.02y_3 + 1.003z_5 - \boxed{v} = -300$$

$$\boxed{x_i} \leq 100$$

$$x_i, y_i, z_i \geq 0$$

Handwritten notes illustrating constraints on x :

- $x = 5$
- $x \leq 5$
- $x \geq 5$
- $-x \leq -5$

Short-term Financing – Sensitivity Report



Variable Cells ✗

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$11	x1	0	-0.003213864	0	0.003213864	1E+30
\$C\$12	x2	0	-3.88578E-16	0	3.88578E-16	1E+30
\$C\$13	x3	0	-0.007118644	0	0.007118644	1E+30
\$C\$14	x4	0	-0.003150847	0	0.003150847	1E+30
\$C\$15	x5	52	0	0	0.003119651	3.80959E-16
\$C\$16	y1	150	0	0	0.003997536	0.003213864
\$C\$17	y2	100	0	0	0.003182044	3.88578E-16
\$C\$18	y3	151.9441675	0	0	3.84731E-16	0.0031603
\$C\$19	z1	0	-0.003997536	0	0.003997536	1E+30
\$C\$20	z2	0	-0.00714	0	0.00714	1E+30
\$C\$21	z3	351.9441675	0	0	0.00393091	0.0031603
\$C\$22	z4	0	-0.003919153	0	0.003919153	1E+30
\$C\$23	z5	0	-0.007	0	0.007	1E+30
\$C\$24	V	92.49694915	0	1	1E+30	1

Constraints ✗

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$J\$11	x1 LHS	150 ↗	-1.037288136 ↓	150	89.17189542	149.4117647
\$J\$12	x2 LHS	100	-1.0302	100	47.05882353	50.98039216
\$J\$13	x3 LHS	-200	-1.02	-200	90.68328348	151.9441675
\$J\$14	x4 LHS	200	-1.016949153	200	90.95533333	152.4
\$J\$15	x5 LHS	-50	-1.01	-50	48	52
\$J\$16	y1 LHS	-300	-1	-300	92.49694915	1E+30
\$J\$19	z1 LHS	0	0	100	1E+30	100
\$J\$20	z2 LHS	0	0	100	1E+30	100
\$J\$21	z3 LHS	0	0	100	1E+30	100
\$J\$22	z4 LHS	0	0	100	1E+30	100
\$J\$23	z5 LHS	52	0	100	1E+30	48

$$-1.037 \times 50$$

Short-term Financing – Questions



- ▶ Assume that Net Cash Flow in January were -200 (instead of - 150). By how much would the company's wealth decrease at the end of June? (Ans = \$ 51.86)
- ▶ Now assume that Net Cash Flow in March were 250 (instead of 200). By how much would the company's wealth increase at the end of June? (Ans = \$ 51.00)
- ▶ Assume that the credit limit were increased from 100 to 200. By how much would the company's wealth increase at the end of June? (Ans = 0)

Non-Linear Optimization

► The LPP deals with linear Objectives and Constraints which are smooth and can be solved fast. Realistically speaking, the Optimization problems we witness involve non-linear terms and are called **non-linear programming problems**. Some examples of non-linear optimization are below

- Economies of Scale (Profits don't grow linearly, but non-linearly due to economies of scale)
- Stochastic Programming (Some coefficients are random variables)
- Minimize Value-at-Risk (VaR is a quantile)
- Mean-Variance Optimization $\max \mu - \frac{1}{2} \sigma^2$
- Creating an Index Fund $\min \text{TF} (w^T \Omega w)$

Univariate Problem – Binary Search

- Suppose we want to find the maximum of

$$f(x) = g'(x) = \boxed{6x^2 - e^x = 0} \quad \begin{matrix} 6 - e \\ = 3.3 \end{matrix} \quad \underline{g(x) = 2x^3 - e^x}$$

$0 - e^0 = -1$

~~Steps~~

(17) ?

To find maximum, we want to solve $f(x) = g'(x) = 6x^2 - e^x = 0$. This is a non-linear equation. So, we want to find a solution to this numerically. For that we need to establish a confidence interval by guess work which contains a solution. We can see $f(x)$ is continuous, and $f(0) = -1$ and $f(1) \approx 3.3$. So, there must be a solution in this interval $[0,1]$

We start with the mid-point $x = 0.5$, find $f(0.5) = -0.15 < 0$,

So, in the next iteration we restrict our confidence interval to $[0.5,1]$. We try out $f(0.75) = 1.258 > 0$

We continue this process. After 20 iterations, the solution converges to 0.5389

- Binary search is fast, and it reduces the interval by a factor of 2 in every iterations
- The downside is that it will converge to only one solution (in case there are multiple local extremums)



A familiar Problem (IRR)

- Suppose we want to find the Internal Rate of Return (IRR) of an investment. It involves solving the following equation. Cashflows each year is given by C_i and the initial investment = V

Internal rate of return (IRR) = $r = ?$

$$V - \frac{C_1}{(1+r)} - \frac{C_2}{(1+r)^2} - \dots - \frac{C_N}{(1+r)^N} = 0$$

$$\frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N}{(1+r)^N} = V$$

A classic use case is YTM of a coupon bond which when used in the above equation in place of r matches the market price of the bond V .

As an example, consider a 4-year non-callable bond with a 10% coupon rate paid annually and a par value of \$1000. If the market price of the bond is \$900, find the YTM ?

$$\frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \frac{100}{(1+r)^3} + \frac{1100}{(1+r)^4} - 900 = 0$$

$$f(r) = 0$$

A familiar Problem (IRR)

Binary Search / Bisection (YTM of a bond)

Bond market Price \$900
 Par \$1,000
 Maturity 4 (yrs)
 Coupon 10%

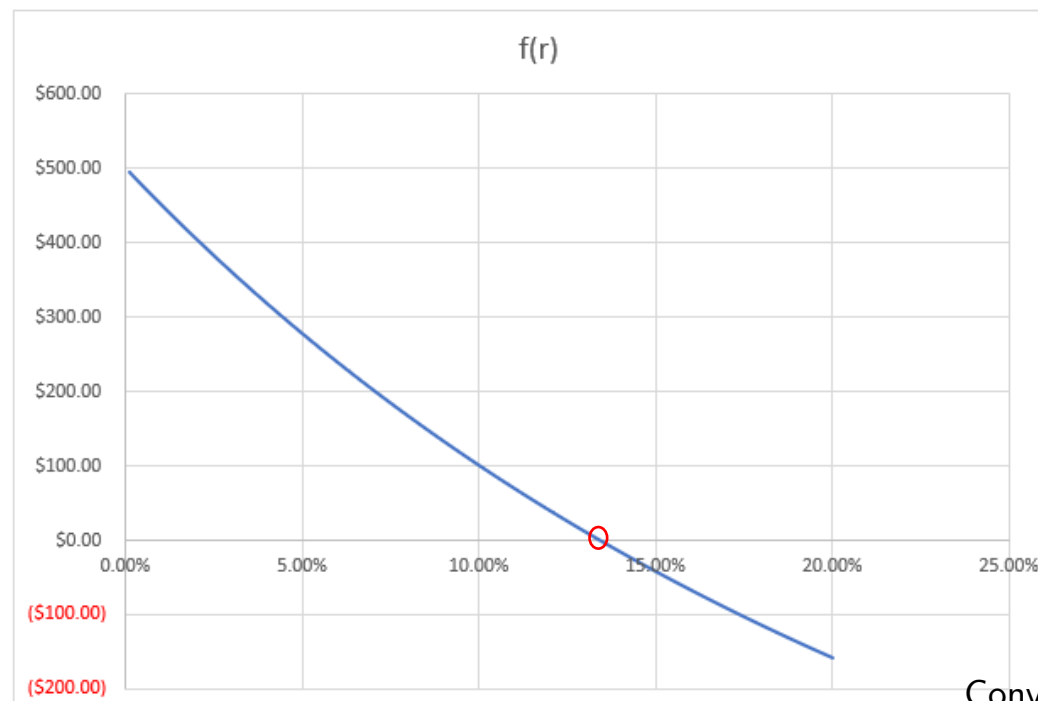
Cashflows

Year	CF
0	\$ (900.00)
1	\$100
2	\$100
3	\$100
4	\$1,100

$$f(r) = \frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \frac{100}{(1+r)^3} + \frac{1100}{(1+r)^4} - 900$$

IRR = 13.389%

r	f(r)
0.10%	\$495.01
0.20%	\$490.05
0.30%	\$485.11
0.40%	\$480.19
0.50%	\$475.30
0.60%	\$470.43
0.70%	\$465.58
0.80%	\$460.76
0.90%	\$455.96
1.00%	\$451.18
1.10%	\$446.42
1.20%	\$441.69
1.30%	\$436.98
1.40%	\$432.29
1.50%	\$427.62
1.60%	\$422.98
1.70%	\$418.36
1.80%	\$413.76



#	LB	UB	r	f(r)
1	10%	15%	0.125	\$24.85902
2	0.125	0.15	0.1375	(\$9.82657)
3	0.125	0.1375	0.13125	\$7.28867
4	0.13125	0.1375	0.13438	(\$1.32492)
5	0.13125	0.13438	0.13281	\$2.96777
6	0.13281	0.13438	0.13359	\$0.81791
7	0.13359	0.13438	0.13398	(\$0.25438)
8	0.13359	0.13398	0.13379	\$0.28154
9	0.13379	0.13398	0.13389	\$0.01352
10	0.13389	0.13398	0.13394	(\$0.12044)
11	0.13389	0.13394	0.13391	(\$0.05346)
12	0.13389	0.13391	0.1339	(\$0.01997)
13	0.13389	0.1339	0.13389	(\$0.00322)
14	0.13389	0.13389	0.13389	\$0.00515
15	0.13389	0.13389	0.13389	\$0.00096
16	0.13389	0.13389	0.13389	(\$0.00113)
17	0.13389	0.13389	0.13389	(\$0.00008)
18	0.13389	0.13389	0.13389	\$0.00044
19	0.13389	0.13389	0.13389	\$0.00018
20	0.13389	0.13389	0.13389	\$0.00005

Convergence

Univariate Problem – Newton's Method

- Another powerful algorithm, referred to as Newton's Method or Newton-Raphson algorithm makes use of the derivative of a function to find the root of the function. Suppose we are interested to find the root of $f(x) = 0$. We start at an initial guess value x_0 . At x_0 the function can be approximated using Taylor Series as

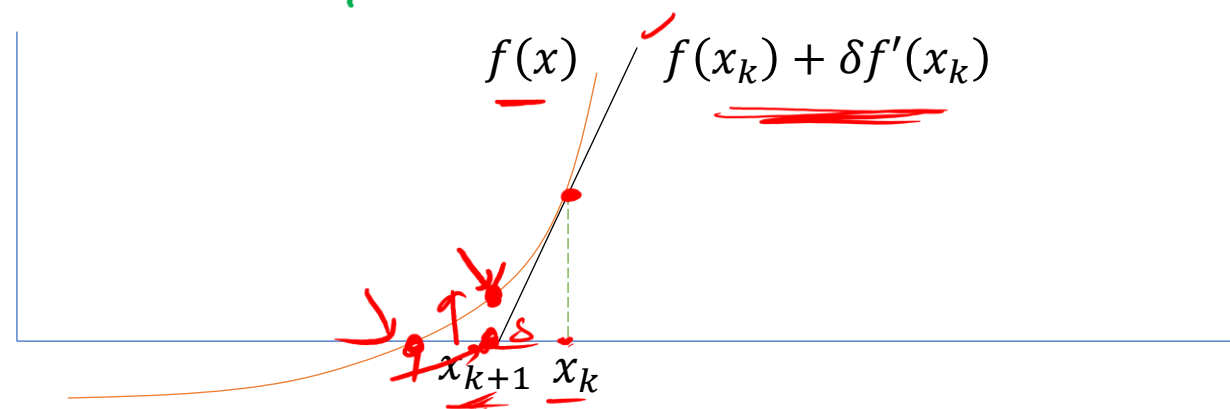
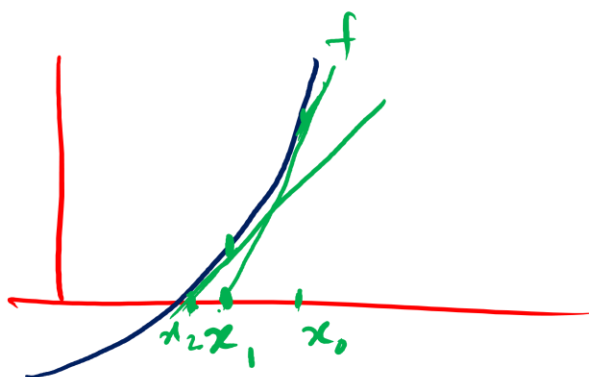
$$f(x_0 + \delta) - f(x_0) \approx \delta \cdot f'(x_0)$$

$$f(x_0 + \delta) \approx f(x_0) + \delta f'(x_0)$$

$$\delta = -\frac{f(x_0)}{f'(x_0)}$$

Setting this to 0 gives us a step in x towards the root. We get $\delta = -\frac{f(x_0)}{f'(x_0)}$

This gives us new value of x : $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$, and we continue this process



Univariate Problem – Newton's Method

- A classic use of Newton Raphson is to solve the implied volatility of Option under Black-Scholes Model. The Black-Scholes Call Option pricing equation is given as

$$\underline{V_{mkt}} = \underline{V(S_0, K, r, \tau, \sigma)} = S_0 N(d_1) - K e^{-rT} N(d_2), \quad d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

The implied volatility is the value of σ that enters in $V(S, K, r, \tau, \sigma)$ such that model price is market price.

$$\underline{f(\sigma)} = \underline{V(\sigma)} - \underline{V_{mkt}} = 0$$

Example – Find the implied volatility of a call option priced at \$ 10.45. Assume initial stock price $S_0 = \$100$, the strike price $K = \$100$, risk free rate $r = 5\%$, time to maturity $T = 1$ yr. We want to find the root of $f(\sigma) = V(\sigma) - V_{mkt} = 0$. The algorithm to find implied volatility takes the form

$$\underline{f'(\sigma)} = \left[\frac{\partial V}{\partial \sigma} \right] = \underline{Vega}$$

$$\underline{\sigma_{k+1}} = \underline{\sigma_k} - \frac{\underline{f(\sigma_k)}}{\underline{f'(\sigma_k)}} = \underline{\sigma_k} - \left(\frac{1}{\underline{vega}} \right) \underline{f(\sigma_k)}$$